

Hardware and Materials Guide

Use Batch 1 only. The public Batch 1 pair is Core rev4.1 plus HAT rev4.1 / R2024.12.06. If the board or packaging says Satellite1.1, rev5.1 Core, rev6.1 HAT, or requires an external Wi-Fi antenna, stop: that hardware is unsupported.

ID	Category	Item	Exact specification	Qty	Required	Where to buy (US)
A01	active driver	Dayton Audio ND91-4	4 ohm full-range driver; manufacturer model ND91-4	1	yes	https://www.parts-express.com/Dayton-Audio-ND91-4-3-1-2-Aluminum-Cone-Full-Range-Driver-4-290-224
A02	passive radiator	Dayton Audio DSA115-PR	4 inch passive radiator with M5 mass post	2	yes	https://www.parts-express.com/Dayton-Audio-DSA115-PR-4-Designer-Series-Aluminum-Cone-Passive-Radiator-295-544
E01	official electronics	FutureProofHomes Satellite1 Batch 1 development kit	Core rev4.1 + HAT rev4.1 / R2024.12.06; not Satellite1.1	1	yes	https://futureproofhomes.net/products/satellite1-kit
H01	insert	CNC Kitchen M3 x 5.7 heat-set insert	M3x0.5 internal thread, 5.7 mm length, 4.6 mm maximum OD	48	yes; includes four spares	https://www.amazon.com/s?k=CNC+Kitchen+M3+heat+set+insert+M3x5.7
H02	speaker cable	2-pin JST-XH 2.54 mm speaker lead	22 AWG stranded red/black, each insulated conductor OD <=1.8 mm, 350 mm minimum	1	yes	https://www.amazon.com/s?k=JST-XH+2.54mm+2+pin+speaker+pigtail

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H03	speaker terminals	solder the leads directly to the driver's terminal lugs	The ND91-4 ships with flat solder lugs, each pierced by an oblong hole: thread the conductor through, crimp it closed, then flow solder, so the joint is mechanically captive rather than relying on solder for strength. Dress the wire toward the driver axis, never outward, and heat-shrink each joint. Radial bulk here is the binding constraint: the terminals are the widest thing that must pass the 76.65 mm bore, they clear it only through the notch, and the leads have to be attached before the driver goes in because the chamber is sealed after.	2 joints	recommended	https://www.amazon.com/s?k=3mm+heat+shrink+tubing+assortment
H04	speaker terminals	2.8 mm fully insulated female quick-disconnects	Alternative to H03, and the weaker one. A disconnect body plus its sleeve stands 2 to 3 mm proud of the lug on the side that has the least room, and it is a rattle source inside a sealed chamber. It also buys nothing: reaching this joint means pulling the driver anyway. If used, verify the blade width against the lug first, measured across the flat face and not along it, and confirm the fit on the speaker-fit coupon before committing to the cabinet print.	2	optional	https://www.amazon.com/s?k=2.8mm+fully+insulated+female+quick+disconnect

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B01	ballast	mild-steel plate	100 x 92 x 6 mm, edges deburred, dry, light oil removed	2	yes	https://www.onlinemetals.com/en/buy/steel/0-25-mild-steel-plate-a36-hot-rolled/pid/1156
B02	radiator tuning	M5 washers plus self-adhesive lead-free strip for final trim	7.99 g per radiator, the two sets matched within 0.02 g, applied centred on the M5 mass post. Stack M5 washers for the bulk -- about 13 of them at roughly 0.6 g each -- then trim self-adhesive lead-free strip for the last fraction of a gram. Weigh both radiators' finished stacks and match them, rather than trusting counts	2 matched sets	yes; final mass requires physical tuning	https://www.amazon.com/s?k=M5+flat+washer+stainless+assortment
G00	gasket stock	closed-cell EPDM foam sheet	2.0 mm nominal, soft, smooth skin, ASTM D1056 2A1 or equivalent	one 300 x 300 mm sheet	yes	https://www.amazon.com/s?k=2mm+closed+cell+EPDM+foam+sheet+adhesive
D01	optional acoustic material	polyester acoustic batting	not installed in RC1; reserve for measurement-led development only	0	no	https://www.parts-express.com/search?keywords=acoustic%20polyfill
O01	official printed part	official_mid_plate	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_mid_plate.stl; preserved official STL	1	yes	printed by you
O02	official printed part	official_mid_plate_threads	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_mid_plate_threads.stl; preserved official STL	1	yes	printed by you
O03	official printed part	official_pcb_spacer	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_pcb_spacer.stl; preserved official STL	1	yes	printed by you

ID	Category	Item	Exact specification	Qty	Required	Where to buy (US)
O04	official printed part	official_lock_ring	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_lock_ring.stl; preserved official STL	1	yes	printed by you
O05	official printed part	official_top_plate	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_top_plate.stl; preserved official STL	1	yes	printed by you
O06	official printed part	official_top_plate_snap_in_diffuser_ring	ASA (PETG alternative); exact file OFFICIAL_PARTS/REQUIRED_SINGLE_MATERIAL/official_top_plate_snap_in_diffuser_ring.stl; preserved official STL	1	yes	printed by you
O07	official printed part	official_top_plate_mm_buttons	ASA (PETG alternative); exact file OFFICIAL_PARTS/OPTIONAL_MULTIMATERIAL/official_top_plate_mm_buttons.stl; preserved official STL	1	optional alternative	printed by you
O08	official printed part	official_top_plate_mm_diffuser_ring	ASA (PETG alternative); exact file OFFICIAL_PARTS/OPTIONAL_MULTIMATERIAL/official_top_plate_mm_diffuser_ring.stl; preserved official STL	1	optional alternative	printed by you
F01	fastener	M3 x 16 shoulder screw	M3 x d4 shoulder screw, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F02	fastener	M3 x 8 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F03	fastener	M3 x 8 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	8	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/

ID	Category	Item	Exact specification	Qty	Required	Where to buy (US)
F04	fastener	M3 x 10 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F05	fastener	M3 x 10 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	8	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F06	fastener	M3 x 6 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F07	fastener	M3 x 10 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F08	fastener	M3 x 8 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F09	fastener	M3 x 8 button head screw	ISO 7380-1, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F10	fastener	M3 x 8 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/
F11	fastener	M3 x 8 socket cap screw	ISO 4762, A2-70 stainless, M3x0.5	4	yes	https://www.mcmaster.com/screws/socket-head-screws/thread-size~m3/

All purchasing availability and prices must be checked by the builder. Manufacturer geometry and electrical parameters are DERIVED_FROM_MANUFACTURER_DRAWING; supplier availability is an ENGINEERING_ESTIMATE.

No structural glue is used. No damping material is installed in RC1. Gaskets are replaceable mechanically compressed EPDM, and the cable seal is TPU 95A.

Source commit 96b3efd90006. No physical validation has been performed.